HKE SOCIETY'S

POOJYA DODDAPPA APPA COLLEGE OF ENGINEERING GULBARGA

DEPARTMENT OF MATHEMATICS

B.E. – I SEM Improvement Test - II

Subject: Calculus and Linear Algebra-I **Course Code: CSMAT101**

Max. Marks: 40 **Date: 01.12.2025** **Time: 9.30 am to 10.45 pm**

NOTE: Answer the following questions.

Q.No:	Questions	Marks
1.a.	Find the rank of matrix by reducing into row echelon form $A = \begin{bmatrix} 0 & 1 & -3 & 1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$	6
1.b.	Test for consistency and solve $x+2y+3z=14,\ \ 4x+5y+7z=35,\ \ 3x+3y+4z=21$	7
1.c.	Solve the following system of equation using Gauss elimination method $x+y+z=9,\ \ x-2y+3z=8,\ \ 2x+y-z=3$	7
2.a.	Let V be the vector space of all real valued continuous functions over R. show that the set W of solutions of differential equations $5\frac{d^2y}{dx^2}-7\frac{dy}{dx}+2y=0$ is a subspace of V	6
2.b	Show that the vector (5,9,5) is a the linear combination of the vectors (2,1,4), (1,-1,3) and (3,2,5)	7
2.c.	Define linear dependence and independence vectors. Check the following vectors for linear independence or dependence. $(i)\{(1,1,-1),(2,-3,5),(-2,1,4)\}$ $(ii)\{(1,3,2),(1,-7,-8),(2,1,-1)\}$	7

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DEPARTMENT OF MATHEMATICS
Major Test - II
NEMERICAL METHODS(CSMAT201)

B.E. – II SEM
TIME: 09:30 AM to 10:45 AM

MAX. MARKS - 40
DATE: 25.05.2026

NOTE: Answer the following questions.

Q.No	Question	Marks
1.a.	A survey conducted in a slum locality reveals the following information as classified below. Income per day(Rs): Under 10 | 10-20 | 20-30 | 30-40 | 40-50 Number of persons: 20 | 45 | 115 | 210 | 115	6
1.b.	Use Lagrange's interpolation formula to find f(4) given x: 0 | 2 | 3 | 6 $f(x)$: -4 | 2 | 14 | 158	7
1.c.	Obtain the piecewise linear interpolation polynomial for the function $f(x)$ defined by the following data x and $f(x)$ x: 1 | 2 | 4 | 8 $f(x)$: 3 | 7 | 21 | 73 hence find $f(3)$ & $f(7)$	7
2.a.	Evaluate $\int_0^{\pi/2} \sqrt{\cos\theta}\, d\theta$ by using Simpson's 3/8th rule	6
2.b.	Evaluate $\int_4^{52} \log_e x dx$ by using Weddle's rule	7
2.c.	Given $y' = x^2 y - 1, y(0) = 1$. Compute $y(0.003)$ by using Taylor's series method	7

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POOJYA DODDAPPA APPA COLLEGE OF ENGINEERING GULBARGA

DEPARTMENT OF MATHEMATICS

Major Test - I

NEMERICAL METHODS (CSMATS21)

B.E. – II SEM | MAX. MARKS - 40

TIME: 09:30 AM to 10:45 AM | DATE: 09.04.2026

NOTE: Answer the following questions.

Q.No	Question	Marks
1.a.	By using Bisection method to find the real root of the equation $x^3 - 2x - 5 = 0$ corrected to three decimal places and carry out five iterations	6
1.b.	By using Regular-Falsi method to find the real root of the equation $\cos x = 3x - 1$ corrected to three decimal places	7
1.c.	By using Newton's-Raphson method to find the real root of the equation $xe^x - 2 = 0$ corrected to three decimal places	7
2.a.	Find the condition number for matrix $A = \begin{bmatrix} 3 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$	6
2.b.	Show that following system of equations ill conditioned $x + y + z = 6,\ x - y + z = 2,\ 2x + y - z = 1$	7
2.c.	Solve the following system of equations by Guass -Seidel method $x + y + 54z = 110,\ 27x + 6y - z = 85,\ 6x + 15y + 2z = 72$	7

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DEPARTMENT OF MATHEMATICS

Major Test - I

Mathematics for Computer science Engg. Stream-II (22MATS21)

B.E. – I SEM | **MAX. MARKS - 40**

TIME: 12:30 PM to 01:45 PM | **DATE: 06.05.2025**

NOTE: Answer the following questions.

Q.No	Question	Marks
1.a.	Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1,-2,-1)$ along $2i-j-2k$	6
1.b.	If $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$ find $div\,\vec{F}$ *and* $curl\,\vec{F}$	7
1.c.	Show that $\vec{F} = (y+z)i + (z+x)j + (x+y)k$ is irrational. Also find scalar function ϕ *such that* $\vec{F} = \nabla\phi$	7
2.a.	Solve $6\frac{d^2y}{dx^2} + 17\frac{dy}{dx} + 12y = e^{-x}$	6
2.b.	Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 5y = e^{2x}\sin x + 2^x$	7
2.c.	Solve $y'' + y = \sec x \tan x$ by the method of variation of parameters	7

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DEPARTMENT OF MATHEMATICS

Major Test - 1

Engg. Mathematics for Computer Science & Engg. Stream-I (22MATS11)

B.E. – I SEM **MAX. MARKS - 40**

TIME: 11:00 AM to 12:15 PM **DATE: 11.11.2024**

NOTE: Answer the following questions.

Q.No	Question	Marks	Course outcome	Cognitive Level
1.a.	With usual notations Prove that $\tan\phi = r\frac{d\theta}{dr}$	6	CO 1	3
1.b.	Find the angle of intersection of the pairs of curves $r = a(1-\cos\theta)$ *and* $r = 2a\cos\theta$	7	CO 1	2
1.c.	Find the pedal equation of the curve $2a/r = (1+\cos\theta)$	7	CO 1	3
2.a.	Find the radius of curvature of the curve $\sqrt{x}+\sqrt{y}=4$ at the point where it cuts the line passing through the origin making an angle 45^0 with the x-axis.	6	CO 1	3
2.b.	Expand $\log(1+\sin x)$ in powers of x by Maclaurin's theorem upto the term containing fourth degree term.	7	CO 2	3
2.c.	Evaluate i) $\lim_{x\to 0}\frac{xe^x - \log(1+x)}{x^2}$ ii) $\lim_{x\to a}\left[2-\frac{x}{a}\right]^{\tan\left(\frac{\pi x}{2a}\right)}$	7	CO 2	3

Course Outcomes	Total Marks Allotted	% Marks Allotted	Cognitive Levels	Total Marks Allotted	% Marks Allotted
CO 1	26	65	L 2	7	18
CO 2	14	35	L3	33	82

B.E. First Semester Examination – January/ February 2026

Differential Calculus and Linear Algebra

Time: 3 hrs] [Maximum Marks: 100

Note: Answer any FIVE full questions, selecting atleast ONE full question from each module.

Module – I

1. a) Obtain the Taylor's expression of $\log_e x$ *about* $x = 1$ up to the term containing fourth degree and obtain $\log_e(1.1)$ (06 Marks)

   b) Obtain the Maclaurin's expression of $\sin^{-1} x$ upto the term containing x^5. (07 Marks)

   c) Evaluate the following limit's (07 Marks)

   i) $\lim_{x\to 0} \dfrac{xe^x - \log(1+x)}{x^2}$ *ii)* $\lim_{x\to 0} \dfrac{1-\cos x}{x\log(1+x)}$

2. a) If $u = \dfrac{1}{\sqrt{x^2+y^2+z^2}}$ then show that $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} + \dfrac{\partial^2 u}{\partial z^2} = 0$ (06 Marks)

   b) If $u + v = e^x \cos y$ *and* $u - v = e^x \sin y$ then find $J = \dfrac{\partial(u,v)}{\partial(x,y)}$. (07 Marks)

   c) Find the extreme values of the function $f(x,y) = x^3 + y^3 - 3x - 12y + 20$ (07 Marks)

Module – II

3. a) Solve $\dfrac{dy}{dx} + y\cot x = \cos x$ (06 Marks)

   b) Solve $\dfrac{dy}{dx} + xy = xy^3$ (07 Marks)

   c) Solve $(x^2 + y^2 + x)dx + xydy = 0$ (07 Marks)

4. a) Solve $D^3 + 6D^2 + 11D + 6y = 0$ (06 Marks)

   b) Solve $6\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} + 12y = e^{-x}$ (07 Marks)

   c) Solve $(D^3 + 1)y = 65\cos(2x+1)$ (07 Marks)

Module – III

5. a) Find the rank of the following matrices by elementary row transformations: (06 Marks)

$$A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$$

   b) Test for consistency and solve $x + y + z = 6,\ x - y + 2z = 5,\ 3x + y + z = 8$ (07 Marks)

   c) Solve the following system of equation using Gauss – Jordan method (07 Marks)
   $x + y + z = 9,\ x - 2y + 3z = 8,\ 2x + y - z = 3$

6. a) Solve the following system of equation by Gauss-elimination method (06 Marks)
   $2x + 5y + 7z = 52;\ 2x + y - z = 0;\ x + y + z = 9$

   b) Find the largest eigen value and corresponding eigen vector of the matrix (07 Marks)

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \text{ with } X^{(0)} = [1,0,0]'$$

c) Show that $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ is diagonalizable. Find an invertible modal matrix P and diagonal matrix D such that $A = PDP^{-1}$. (07 Marks)

Module – IV

7. a) Let V be the vector space of all valued continuous function over R. Show that the set W of solution of differential equations $5\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 2y = 0$ is a subspace of V. (06 Marks)

b) Show that the vector, (5,9,5) is a linear combination of the vectors (2,1,4), (1,-1,3) and (3,2,5). (07 Marks)

c) Given the set $S = \{(0,2,3,4)\ (2,3,5,4)\ (4,8,13,12)\}$ show that S is linearly dependent. (07 Marks)

8. a) Find the basis and dimension of the subspace of V spanned by the following vectors $\{(1,2,3,2),(2,3,5,1),(1,3,4,5)\}$ (06 Marks)

b) Show that the following vectors are mutually orthogonal
$a = (2,1,3)$; $b = (-6/7,15/14,3/14)$; $c = (1/3,1/3,-1/3)$ (07 Marks)

c) Show that the following pairs of vectors are orthonormal
$a = (1,0,0,0)$; $b = \left(0,1/\sqrt{5},2/\sqrt{5},0\right)$ (07 Marks)

Module – V

9. a) Show that the transformation $T : V_3(R) \to V_2(R)$ defined by $T(x,y,z) = (y-x, y-z)$ is a linear transformation. (06 Marks)

b) Find the matrix of linear transformation $T : V_2(R) \to V_2(R)$ defined by $T(x,y) = (x-y)$ relative to the basis is $X = \{e_1, e_2\}$ *and* $Y = \{(1,1),(1,-1)\}$ (07 Marks)

c) For the matrix $A = \begin{bmatrix} 4 & -2 & 1 \\ 0 & 5 & 3 \\ 2 & -1 & 1 \end{bmatrix}$ determine whether it is singular or non-singular. (07 Marks)

10. a) $S : R^2 \to R^2$ be the linear transformation represented by the matrix $B = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$

i) Determine whether S is singular or non-singular (06 Marks)

ii) If non-singular, compute B^{-1} and verify $BB^{-1} = I_2$

b) Find the range, null space, rank, nullity in the case of the following linear transformations. Also verify the rank nullity theorem.
$T : V_3(R) \to V_2(R)$ *define by* $T(x,y,z) = (y-x, y-z)$ (07 Marks)

c) State Rank nullity theorem and for the linear transformation
$T : R^4 \to R^3$, $T(x_1,x_2,x_3,x_4) = (x_1+x_2, x_2+x_3, x_3+x_4)$
i) Find the matrix of T ii) Determine the rank and nullilty and iii) Verify rank-nullity (07 Marks)

B.E. Second Semester Examination – June/July 2026
Numerical Methods

Time: 3 hrs] [Maximum Marks: 100

Note: Answer any FIVE full questions, selecting atleast ONE full question from each module.

Module – I

1. a) Define the following i) Inherent error ii) Rounding off error iii) Truncation error iv) Absolute error v) Relative error vi) Percentage error (06 Marks)

   b) By using bisection method to find the real root of the following equations corrected to three decimal places and carry out five iterations $x^3 - 4x - 9 = 0$ (07 Marks)

   c) Use secant method to find a real root of the equation $x^3 - 5x - 1 = 0$ (07 Marks)

2. a) Find the real root of the equation $\cos x = 3x - 1$ correct to three decimals using regular falsi method. (06 Marks)

   b) Compute the real root of $x \log_{10} x - 1.2 = 0$ by the method of falsi position – carry out three iterations. (07 Marks)

   c) Use Newton-Raphson method to find $\sqrt[3]{37}$ correct to 3 decimal places. (07 Marks)

Module – II

3. a) If $V = (0,-1,1,2)$ and $W = (1,5,1,8)$ compute L_1, L_2, L_∞ norms of V, W and $2V + W$. (06 Marks)

   b) Find the condition number of the following matrix (07 Marks)

$$\begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$$

   c) Show that following system of equations ill conditioned $x + y + z = 6$, $x - y + z = 2$, $2x + y - z = 1$ (07 Marks)

4. a) Solve the following system of equations by Gausses-Seidel method to obtain the final solution correct to three places of decimal (06 Marks)
   $x + y + 54z = 110$, $27x + 6y - z = 85$, $6x + 15y + 2z = 72$

   b) Find the largest eigen value and the corresponding eigen vector of the matrix A by the power method given that (07 Marks)

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

   c) Compute the eigen values and the eigen vectors of the following matrices by applying Jacobi's method (07 Marks)

$$A = \begin{bmatrix} 6 & \sqrt{3} \\ \sqrt{3} & 4 \end{bmatrix}$$

Module – III

5. a) Find $u_{0.5}$ from the data u_0=225, u_1= 238, u_2 = 320, u_3 = 340, using appropriate interpolation formula. (06 Marks)

   b) A survey conducted in a slum locality reveals the following information as classified below

Income per day (R.S)	Under 10	10-20	20-30	30-40	40-50
Number of persons	20	45	115	210	115

   Estimate the probable number of persons in the income group 20 to 25 using appropriate interpolation formula. (07 Marks)

c) Use Newton's divided difference formula to find f(4) given the data. (07 Marks)

x	0	2	3	6
f(x)	-4	2	14	158

6. a) Use Langrange's interpolation formula to find y at x = 10 given. (06 Marks)

x	5	6	9	11
y	12	13	14	16

b) Obtain the piecewise linear interpolation polynomial for the function f(x) defined by following data x and f(x) find f(3) and f(7) (07 Marks)

x	1	2	4	8
f(x)	3	7	21	73

c) Obtain the piecewise quadratic interpolating polynomial defined by the following data (07 Marks)

x	-2	-1	1	2	4
f(x)	-29	-8	-2	-5	7

Hence find f(- 0.5) and f(2)

Module – IV

7. a) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Simpson's $3/8^{th}$ rule taking four equal strips and hence deduce an appropriate value of π.. (06 Marks)

b) Evaluate $\int_4^{5.2} \log_e x$ taking 6 equal strips by applying Weddle's rule. (07 Marks)

c) Use Taylor's series method to solve the following initial value problem. (07 Marks)

$\frac{dy}{dx} = x^2 + y^2,\ y(0) = 1$ to find y at x = 0.1, x = 0.2, x = 0.3

8. a) Using modified Euler's method to find y at x = 0.2, given $\frac{dy}{dx} = 3x + \frac{1}{2}y$ with y(0) = 1 (06 Marks)

taking h = 0.1. perform three iterations at each step.

b) Use fourth order Runge-Kutta method to find y at x = 0.6 given (07 Marks)

$(x+y)\frac{dy}{dx} = 1,\ y(0.4) = 1,$ correct to four decimal places.

c) Apply Milne's method to compute y(1.4) correct to four decimal places given (07 Marks)

$\frac{dy}{dx} = x^2 + \frac{y}{2}$ and data $y(1) = 2,\ y(1.1) = 2.2156,\ y(1.2) = 2.4649,\ y(1.3) = 2.7514$

Module – V

9. a) Find the directional derivative of function $\phi = 4xz^3 - 3x^2y^2z$ $at(2,-1,2)$ along $2i - 3j + 6k$ (06 Marks)

b) Find $div\ \vec{F}$ and $curl\ \vec{F}$, where $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$ (07 Marks)

c) Show that $\vec{F} = (y+z)i + (z+x)j + (x+y)k$ is irrotational. Also find a scalar function ϕ such that $\vec{F} = \nabla\phi$ (07 Marks)

10. a) Express the vector $\vec{A} = 2xi - 3y^2j + xzk$ in the cylindrical polar co-ordinate system. (06 Marks)

b) Obtain the Jacobian of the cylindrical system. (07 Marks)

c) Represent $\vec{F} = yi - zj + xk$ in spherical polar co-ordinates and hence find F_r, F_θ, F_ϕ. (07 Marks)

B.E. Second Semester Examination – September/October 2025
(Summer Term)

Mathematics for CSE Stream - II

Time: 3 hrs] [Maximum Marks: 100

Note: Answer any FIVE full questions, selecting atleast ONE full question from each module.

Module – I

1. a) Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point (2, -1 ,2) **(06 Marks)**

   b) Find the directional derivative of $\phi(x, y, z) = x^2yz^3$ at (2, 1, -1) in the direction $-4i - 4j + 12k$ **(07 Marks)**

   c) Show that $\vec{F} = (2xy^2 + yz)i + (2x^2y + xz + 2yz^2)j + (2y^2z + xy)k$ is a conservation force field. Find its scalar potential. **(07 Marks)**

2. a) Find the Scalar factors of cylindrical system. **(06 Marks)**
   b) Show that co-ordinates of spherical system are orthogonal. **(07 Marks)**
   c) Find expression for $curl\ \vec{A}$ in orthogonal curvilinear coordinates. **(07 Marks)**

Module – II

3. a) Solve: $(4D^4 - 4D^3 - 23D^2 + 12D + 36)y = 0$ **(06 Marks)**

   b) Solve : $y'' - 6y' - 9y = 5e^{-2x}$ **(07 Marks)**

   c) Apply the method of variation of parameters to solve : $y'' + 4y = 4\sec^2 2x$ **(07 Marks)**

4. a) Solve $(1+x)^2 \frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + y = \sin 2[\log(1+x)]$ **(06 Marks)**

   b) Solve $x^2 \frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = x^2 \log x$ **(07 Marks)**

   c) Solve $(D^4 + 8D^2 + 16)y = 2\cos^2 x$ **(07 Marks)**

Module – III

5. a) Define vector space with example.
   b) Find whether the set of vectors $\{(4,0,2,1)(2,1,3,4)(2,3,4,7)(2,3,1,4)\}$ are linear independent? **(06 Marks)** **(07 Marks)**

   c) Express the polynomial $f(x) = x^2 + 4x - 3$ as a linear combination of the $f_1(x) = x^2 - 2x + 5$, $f_2(x) = 2x^2 - 3x$ *and* $f_3(x) = x + 3$ **(07 Marks)**

6. a) Let $T : V_3(R) \to V_2(R)$ be defined as $T(x, y, z) = (x + y, y + z)$. Find the matrix T w.r.t the basis, $\{(1,1,1)(1,0,0)(1,1,0)\}$ and also w.r.t standard basis. **(06 Marks)**

   b) Find the linear transformation $T : R^2 \to R^2$ satisfying $T(1,1) = (0,1)$ *and* $T(-1,1) = (3,2)$ **(07 Marks)**

   c) Find range, rank, nullify for the following $T : V_3(R) \to V_3(R)$ defined by $T(x, y, z) = (x + y, x - y, 2x + z)$ **(07 Marks)**

Module – IV

7. a) Apply Newton Raphson's method to find an approximation root of the equation $x^3 - 3x - 5 = 0$ which lies near 2. Carry out 3 iterations. (06 Marks)

b) Find f(0.1) using Newton's forward interpolation formula, for the following (07 Marks)

x	0	1	2	3
f(x)	1	3	7	13

c) Construct the interpolation polynomial using Newton's divided difference formula (07 Marks)

x	2	4	5	6	8	10
f(x)	10	96	196	350	868	1746

8. a) Compute real root of $x \log_{10} x - 1.2 = 0$ by Regula-Falsi method (06 Marks)

b) Given: f(40) = 184, f(50) = 204, f(60) = 226, f(70) = 250, f(80) = 276, f(90) = 304. Find f(38) and f(85) using suitable interpolation formulae. (07 Marks)

c) Use Langranges interpolation formula to find f(4) given. (07 Marks)

x	0	2	3	6
f(x)	-4	2	14	158

Module – V

9. a) Use Taylor's series method to find y at x = 0.1, given $\frac{dy}{dx} = x^2 + y^2$, $y(0) = 1$ (06 Marks)

b) Evaluate $\int_2^8 \frac{dx}{\log_{10} x}$ by using Simpson's 1/3 rule. (07 Marks)

c) Given $\frac{dy}{dx} = 3x + \frac{1}{2}y$. $y(0) = 1$ compute y(0.2) by taking h = 0.2 using R-K method of fourth order. (07 Marks)

10. a) Evaluate $\int_0^1 \frac{dx}{1+x}$ by using Simpson's $3/8^{th}$ rule. Hence deduce the value of $\log_e 2$. (06 Marks)

b) Apply modified Euler's method to find f(0.2) correct to four decimal places given that $\frac{dy}{dx} = x - y^2$, $y(0) = 1$ taking h = 0.1 (07 Marks)

c) Apply Milne's method to compute y(1.4) given $\frac{dy}{dx} = x^2 + \frac{y}{2}$ and data $y(1) = 2$, $y(1.1) = 2.2156$, $y(1.2) = 2.4649$, $y(1.3) = 2.7514$ (07 Marks)

B.E. First Semester Examination – January 2025

Mathematics for CSE Stream - I

Time: 3 hrs] [Maximum Marks: 100

Note: Answer any FIVE full questions, selecting atleast ONE full question from each module.

Module – I

1. a) With usual notations prove that $\tan\varphi = r\frac{d\theta}{dr}$ **(06 Marks)**

   b) Find the angle between the pairs of curves $r = a\log\theta$ *and* $r = a/\log\theta$ **(07 Marks)**

   c) Find the pedal equation for the curve $r^m = a^m(\cos m\theta + \sin m\theta)$ **(07 Marks)**

2. a) Show that the radius of curvature of the centenary of uniform strength $y = a\log\sec(x/a)$ **(06 Marks)**

   b) Derive an expression for the radius of curvature for polar form. **(07 Marks)**

   c) Write a program for angle between the curves $r = 4(1+\cos\theta)$ and $r = 5(1-\cos\theta)$ **(07 Marks)**

Module – II

3. a) Obtain the Maclaurin's expansion of the function log(1+x) and hence deduce that $\log\sqrt{\frac{1+x}{1-x}} = x + \frac{x^3}{3} + \frac{x^5}{5} + \ldots..$ **(06 Marks)**

   b) Evaluate *i)* $\lim_{x\to 0}\frac{1-\cos x}{x\log(1+x)}$ *ii)* $\lim_{x\to a}\left[2-\left(\frac{x}{a}\right)\right]^{\tan\frac{\pi x}{2a}}$ **(07 Marks)**

   c) If $u = \log\sqrt{x^2+y^2+z^2}$ show that $(x^2+y^2+z^2)\left[\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\right] = 1$ **(07 Marks)**

4. a) If $u = f\left(\frac{x}{y},\frac{y}{z},\frac{z}{x}\right)$, show that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z} = 0$ **(06 Marks)**

   b) If $x = r\sin\theta\cos\phi, y = r\sin\theta\sin\phi, z = r\cos\theta,$ show that $\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = r^2\sin\theta$ **(07 Marks)**

   c) Find the extreme values of the function $f(x,y) = x^3+y^3-3x-12y+20$. **(07 Marks)**

Module – III

5. a) Solve $\frac{dy}{dx}+y\cot x = \cos x.$ **(06 Marks)**

   b) Solve $\frac{dy}{dx}+\frac{(y\cos x+\sin y+y)}{\sin x+x\cos y+x} = 0$ **(07 Marks)**

   c) Find the orthogonal trajectories of the curve $r = a(1-\cos\theta)$ **(07 Marks)**

6. a) Solve $p^2+2py\cot x = y^2$ **(06 Marks)**

   b) Solve the equation *(Px-y)(Py+x)* = 2P by reducing into clairauts form **(07 Marks)**

   c) Write a program to solve $y''-5y'+6y = \cos 4x$ **(07 Marks)**

Module – IV

7. a) Evaluate $\int_0^1 \int_x^{\sqrt{x}} (x^2 + y^2)\, dy\, dx$ (06 Marks)

b) Evaluate $\iint_R xy\, dx\, dy$ where R is the region bounded by the coordinates axes and the line $x+y = 1$. (07 Marks)

c) Change the order of integration and evaluate $\int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y}\, dy\, dx$ (07 Marks)

8. a) Prove the relation $\beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ (06 Marks)

b) Evaluate $\int_0^{\infty} \frac{dx}{1+x^4}$ (07 Marks)

c) Prove that $\int_0^{\pi/2} \sqrt{\sin\theta}\, d\theta \times \int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin\theta}} = \pi$ (07 Marks)

Module – V

9. a) Find the rank of the matrix

$$A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$ (06 Marks)

b) Test for consistency and solve $x - y + z = 6,\ x - y + 2z = 5,\ 3x + y + z = 8$ (07 Marks)

c) Solve the following system of equation using Gauss – Jordan method
$x + y + z = 9,\ x - 2y + 3z = 8,\ 2x + y - z = 3$ (07 Marks)

10. a) Solve the following system of equation by using Gauss seidal method
$20x + y - 2z = 17,\ 3x + 20y - z = -18,\ 2x - 3y + 20z = 25$ (06 Marks)

b) Find the eigen value and eigen vector of the matrix

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$ (07 Marks)

c) Find the largest eigen value and corresponding eigen vector of the matrix A by the power method $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ with $[1,0,0]^1$. (07 Marks)